

Product Brief: Diagnostic Structurel IA (*working title* — see *Open Questions*)

Working document, in English. **The product itself ships in French** (hard requirement).

[ASSUMPTION] tags mark anything inferred — correct them.

Executive Summary

Diagnostic Structurel IA is a mobile-first, offline-capable field tool for structural-diagnostic experts. It guides the on-site survey, uses AI to pre-identify structural disorders (*désordres*) from site photos, and auto-assembles the compliant PDF deliverable — technical cross-sections (*coupes*), annotated schematics (*schémas*), and disorder localization included. The insight driving it is simple and lived: **for a practicing diagnostician, the site visit is the fast part**. A full diagnostic takes ~3–3.5 days, yet only ~0.5 day is on-site. The other ~2.5–3 days is desk work — writing the report, placing annotated photos, drawing coupes and schémas. This product attacks the deskwork, not the inspection.

Two things make now the moment. France's **Décret n°2025-814 (Aug 2025)**, under the *loi Habitat dégradé*, makes structural diagnosis of collective housing mandatory in designated zones — raising demand against an unchanged manual bottleneck. And computer-vision defect detection is now mature enough to reliably *assist* expert judgment. Critically, French regulation requires a **Bac+5 engineer to sign every diagnosis** and carry **≥€1M professional liability** — so the AI can never diagnose on its own. This product is built around that reality: **the AI proposes, the expert disposes and signs**. That is not a limitation to hedge; it is the only legally and professionally honest model.

It is being built by a practicing structural-diagnostic expert who is also a **data engineer**, with a second expert as pilot — **for his own daily use first, and as a subscription product for other francophone diagnosticians second**. That dual profile matters: he holds the crown-jewel inputs (his own site photos, past reports, worked examples, and reference norms) needed to seed the AI, and he is his own first user and first labeler.

The Problem

Structural diagnosis today rests on expert judgment, manual site surveys, and empirical interpretation — robust, but slow, variable between operators, and hard to capitalize at scale. For a solo expert the cost is concrete and personal:

- A single diagnostic consumes **~3–3.5 days**. Only **~0.5 day is on-site**; the rest is desk work.
- **The report write-up alone is ~1.5 days** — the part most dreaded and least reusable.
- Placing annotated photos against the right disorders: **~0.5 day**. Drawing technical coupes and schémas: **~0.5–1 day**.

This deskwork caps throughput: fewer diagnostics per month, slower turnaround, and consistency that depends on the operator's day. Meanwhile new French regulation is set to raise the volume of mandatory diagnoses — **more demand hitting the same manual ceiling**. Generic field-inspection apps don't relieve this: they manage snags and photos, but none produce the *compliant structural deliverable* — coupes, schémas, disorder mapping, regulatory layout — that the job actually requires.

The Solution

Diagnostic Structural IA compresses the ~2.5–3 days of deskwork into a **review-and-sign step**.

- **On site, offline:** guided dynamic survey by structure type, checklists, intensity sliders (1–5), geolocated photos pinned to structural elements — no network needed, no data lost.
- **AI-assisted reading:** submitted photos are pre-classified by disorder type (cracking, corrosion/carbonation, moisture/infiltration, deformation, masonry, misc.) with a severity indication and a confidence score. The expert **validates, corrects, or rejects** each detection in seconds — and adds what the model missed.
- **Compliant deliverable, generated:** the app assembles the PDF — cover page, TOC, per-zone synthesis, annotated photos, ≥ 1 relevant coupe, an annotated schéma, and automatic disorder localization consistent with the survey numbering — in one of several configurable report templates.
- **A learning loop:** every validated correction is stored as training data and exported to periodically retrain the model, with a dashboard tracking precision/recall by disorder category.

The outcome the expert feels: **the same diagnostic, minus two days at a desk**.

What Makes This Different

The defensible edge is **not** the AI. Crack detection is ~99% on open benchmarks and commoditizing; the honest in-the-wild target here is **>75% accuracy on cracks and corrosion, always expert-validated**. The real edge is elsewhere:

- **Deep fit to the actual French diagnostic workflow and its compliant deliverable** — the coupes, schémas, disorder mapping, and regulatory report structure that generic tools (e.g. **PlanRadar**, the EU field-inspection incumbent) and enterprise/drone players (**STRUCINSPECT**, **Niricson**) do not serve at the level of the individual on-foot diagnostician.
- **A founder who is both the domain expert and the data engineer** — building for a workflow he lives daily, with his own labeled data to seed the model and beat the cold start that stops most entrants.
- **Francophone + regulatory alignment as a first-class design goal**, not an afterthought.

Two things we deliberately **will not** claim as moats: the CV model itself (commoditized), and the learning loop as "defensibility" (a genuine quality *flywheel*, but incumbents hold more and better-structured data — we frame it as continuous improvement, not a wall). Honest beats impressive here.

Who This Serves

- **Primary (v1): the founding expert himself**, plus his supporting expert as pilot — solo/duo structural diagnosticians who run the full mission end-to-end. Success for them: reclaim the deskwork days, do more diagnostics, and produce consistent, defensible, regulation-ready reports every time.
- **Next (commercial):** other independent structural diagnosticians, small *cabinets d'expertise*, and *bureaux d'études* in francophone markets facing the same bottleneck and rising regulatory demand.

Three in-product roles carry across: **field expert** (surveys, photos, AI validation, schémas, PDF), **project lead** (creates missions, assigns experts, consults reports), **administrator** (users, settings, AI-model monitoring, report templates).

Success Criteria

Product acceptance (from the working spec §9 — full list in [addendum.md](#)):

- Offline: create a mission, complete a full survey, capture photos with no network; sync with **zero data loss within 30s** of reconnection.
- **Survey entry $\geq 30\%$ faster** than the current manual process, validated by pilot experts.
- Generates a **compliant PDF**: all survey data, annotated photos, ≥ 1 relevant coupe, an annotated schéma, disorder mapping consistent with survey numbering, correct regulatory layout.
- AI: **$>75\%$ accuracy on cracks and corrosion** on the test set; confidence score per detection; validate/correct/reject each in **$<5s$** ; corrections exportable for retraining.
- iOS + Android on the 5 most common expert devices; no data loss on force-close; an untrained expert productive in **<1 hour**.

Personal ROI (the reason it exists): **[ASSUMPTION]** cut per-diagnostic turnaround from ~ 3.5 days toward **~ 1.5 days** — the ~ 2.5 days of deskwork is the target — enabling roughly **$2\times$ throughput**. *Confirm your real target.*

Commercial signal (gate before productizing): **[ASSUMPTION]** after a pilot, a handful of other experts express willingness to pay; pricing model (per-month vs per-report), price point, and target segment **TBD**.

Scope

In (v1):

- **Structure types: buildings** — collective housing, commercial/ERP, industrial.
- **Module 1** — guided offline field surveys + photo capture/annotation + compliant PDF generation (coupes, schémas, disorder localization, configurable templates).
- **Module 2** — AI disorder detection/classification for the main categories, with severity, confidence, expert validation.
- **Module 3** — structured disorder database + learning loop + AI performance dashboard.
- Platforms: iOS + Android. **UI language: French** (hard requirement). Offline-first.

Out (v1):

- **Ouvrages d'art** (bridges, tunnels) — deferred to Vision.
- Mission accounting/billing; automated legal-expertise report authoring (per spec exclusions).
- Multi-tenant subscription / commercial rollout — **gated behind personal validation**.

Vision

If it works for one expert, it works for many. In **2–3 years**, Diagnostic Structurel IA graduates from a personal tool to **the reference field-and-report platform for francophone structural diagnosticians** — extended to *ouvrages d'art*, sold as a subscription, with an accumulated corpus of expert-validated diagnoses that compounds into a real quality advantage (honestly: a flywheel earned over time, not a claimed moat on day one).

The near-term proof gate is unglamorous and specific: **does it cut the founder's own deskwork in half, on real missions, this year?** Everything else follows from that.

Open Questions / Assumptions to Firm Up

- **Product name/brand** — "Diagnostic Structurel IA" is a working title.
- **Personal ROI target** — the real turnaround/throughput numbers you're aiming for.
- **Subscription** — pricing model, price point, target segment, and a concrete willingness-to-pay signal.
- **Timeline & budget** for the build (you're technical + one pilot expert — solo/duo build assumed).
- **Geographic/market focus** — [ASSUMPTION] France / francophone first.